

NETFLIX

NETFLIX RECOMMENDATION SYSTEM

DYNAMIC HYBRID SVD RECOMMENDER

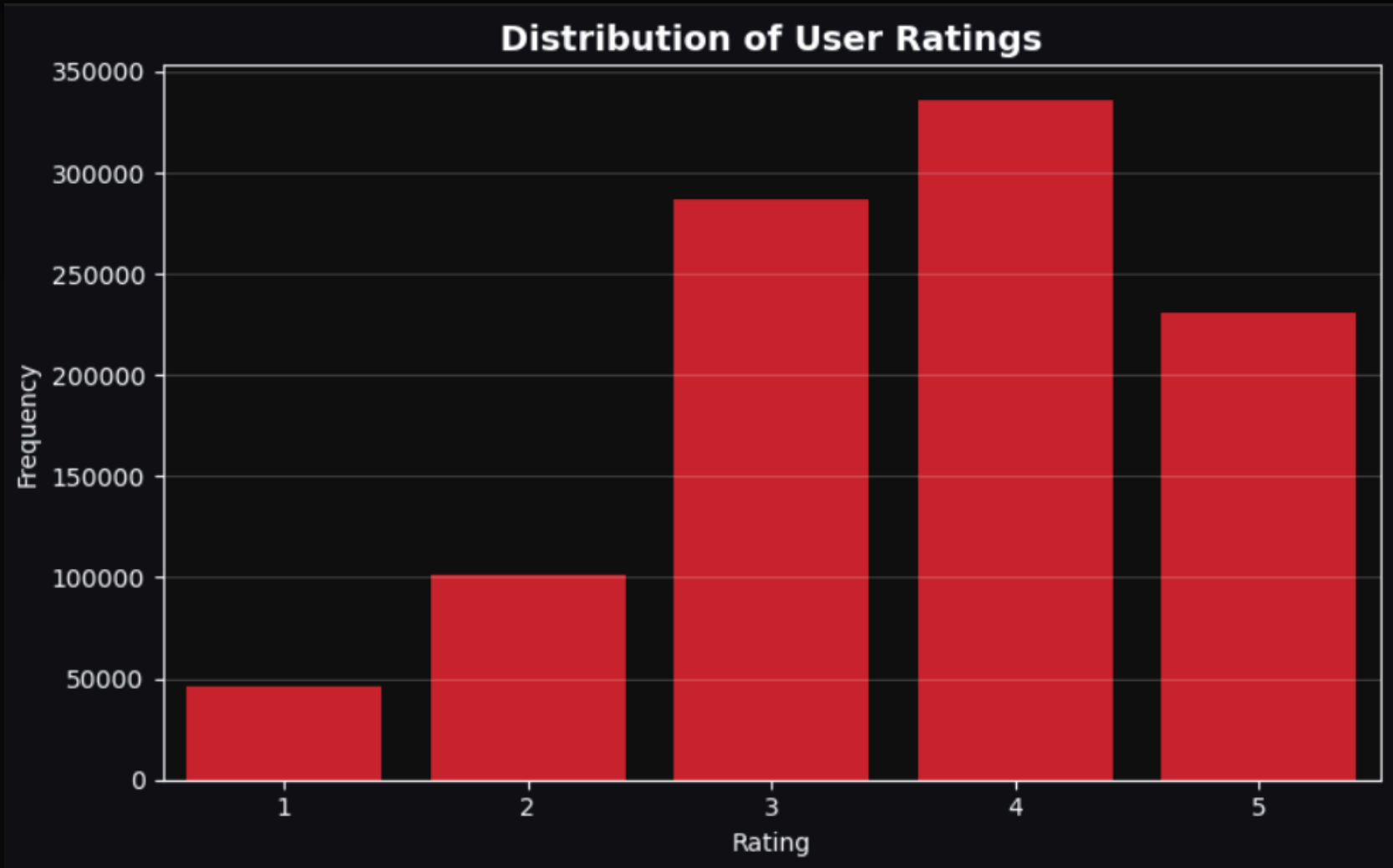
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PROBLEM OVERVIEW & DATASET

PROBLEM STATEMENT

-  **Content Overload**
Streaming platforms have thousands of movies, making content discovery difficult.
-  **Lack of Personalization**
Users receive generic recommendations that fail to reflect individual preferences.
-  **Cold-Start Challenge**
Users with limited rating history are difficult to recommend accurately.
-  **Need for Intelligent Recommendations**
The system should recommend relevant movies while effectively handling cold-start users.



DATASET SUMMARY



999,994
ratings



289,387
users



16,144
movies



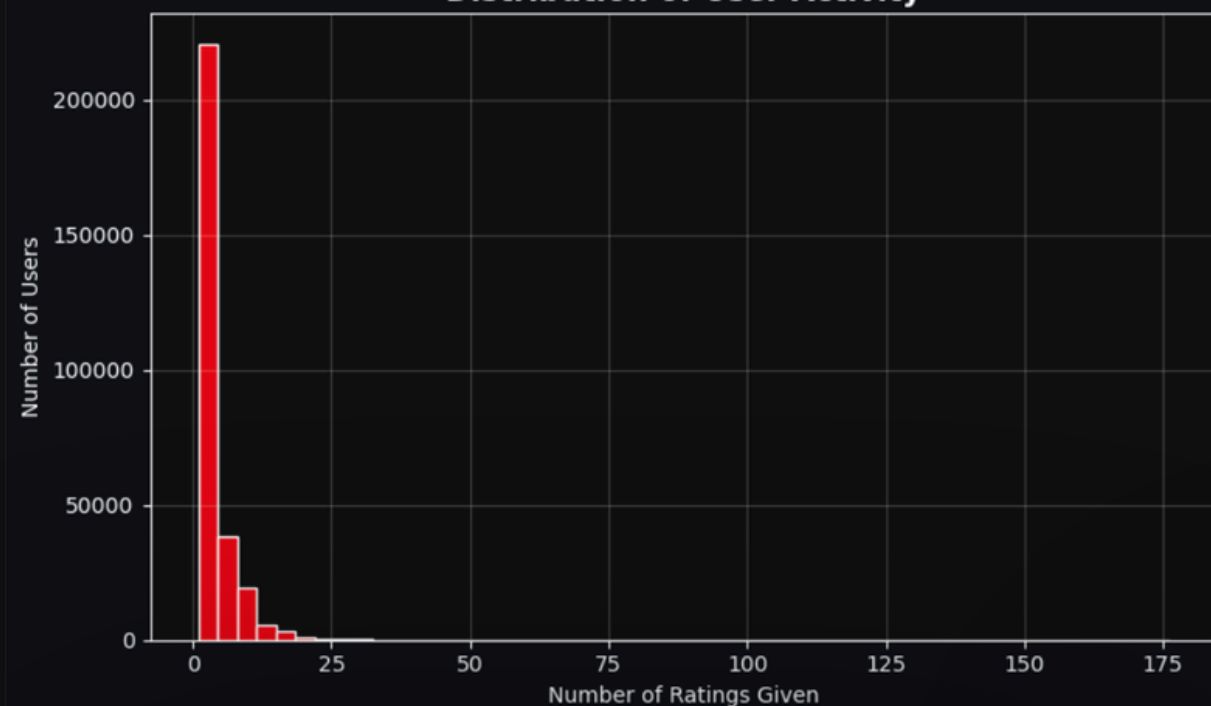
3.61
avg rating

 Sparsity level: 99.9786%

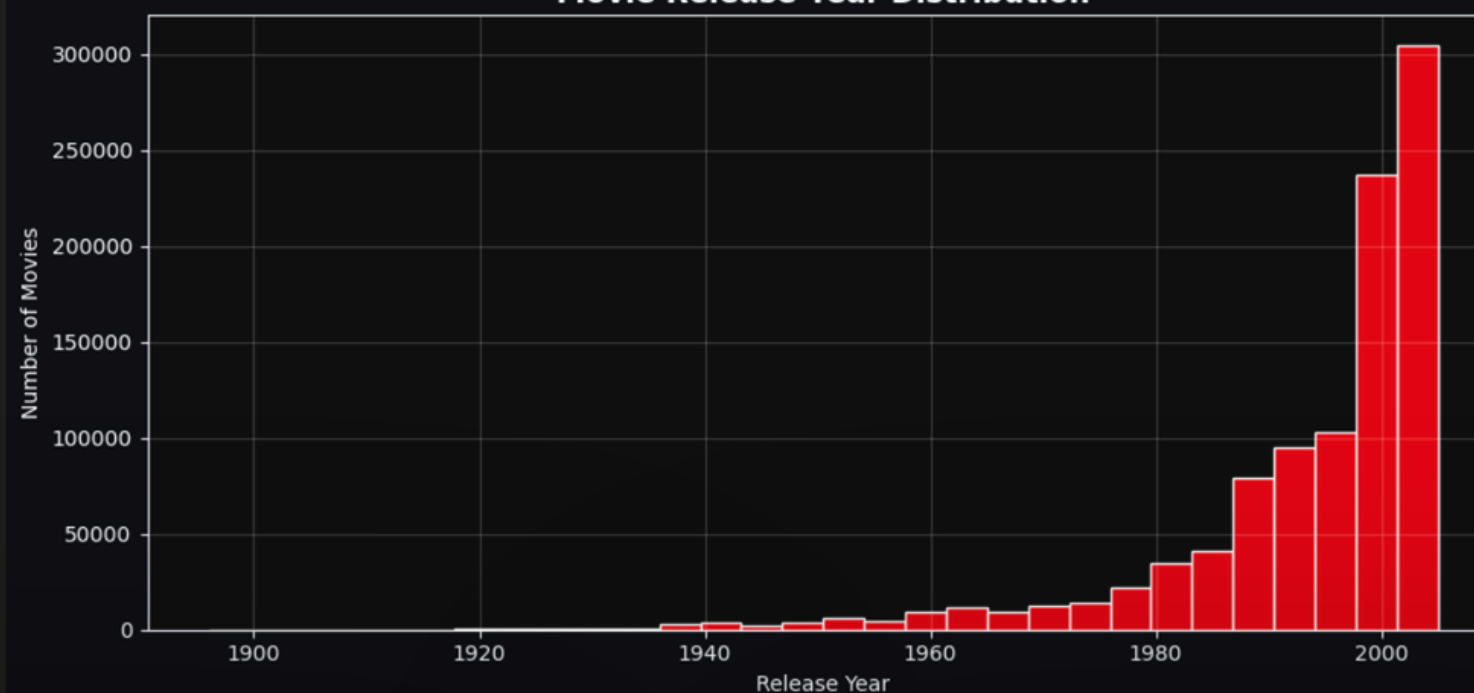
KEY INSIGHTS

- ★ Most ratings lie between 3 and 5
- ❄ 68.05% users are cold-start users
- 🎯 Personalization is challenging due to sparse interactions

Distribution of User Activity



Movie Release Year Distribution



Highly Skewed User Activity

A small fraction of users contribute most ratings, while the majority interact with only a few movies.



Modern Content Dominance

Most movies were released between 1980–2005, indicating a strong focus on contemporary content.



Positive rating bias

Ratings are concentrated towards the higher end, suggesting users mainly rate movies they enjoy



HOW THESE FINDINGS INFLUENCED MODEL DESIGN

- User behavior is highly uneven, making it difficult to learn preferences for many users.
- Sparse interactions reduce the effectiveness of traditional similarity-based recommenders.
- Positive rating bias requires models capable of capturing subtle preference differences.
- These findings motivated the use of Matrix Factorization (SVD) as the foundation of the recommendation system.



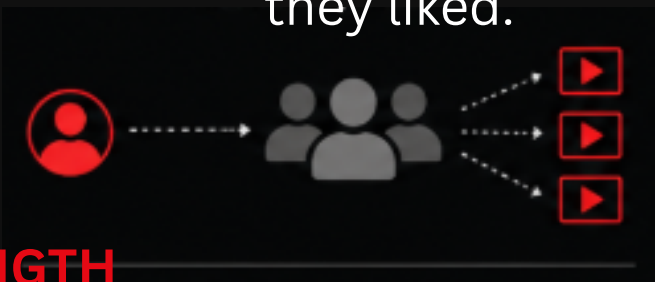


OUR APPROACH

We implemented and compared three collaborative filtering approaches to identify the most effective model for movie recommendations.

1. USER-BASED CF

Find similar users and recommend movies they liked.



STRENGTH

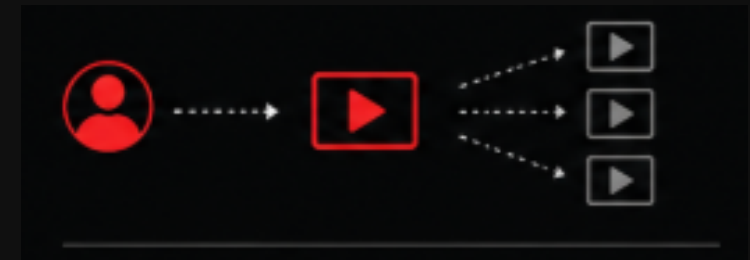
Captures user preferences from like-minded users.

LIMITATION

Struggles with scalability and sparse user histories.

2. ITEM-BASED CF

Find similar movies and recommend those similar to what the user liked.



STRENGTH

More scalable and effective with large user bases.

LIMITATION

Depends on item co-occurrence in interaction data.

3. SVD MATRIX FACTORIZATION

Learn latent features for users and movies in a low-dimensional space



STRENGTH

Handles sparsity well and learns hidden preferences.

LIMITATION

Computationally intensive for very large datasets.

WHY SVD WAS CHOSEN

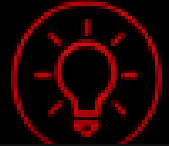


- Achieved the best RMSE among all models.
- Learns latent user and movie factors, capturing deeper preferences.
- Performs better on sparse datasets and generalizes well to unseen data.
- Selected as the base model for the Dynamic Hybrid SVD approach



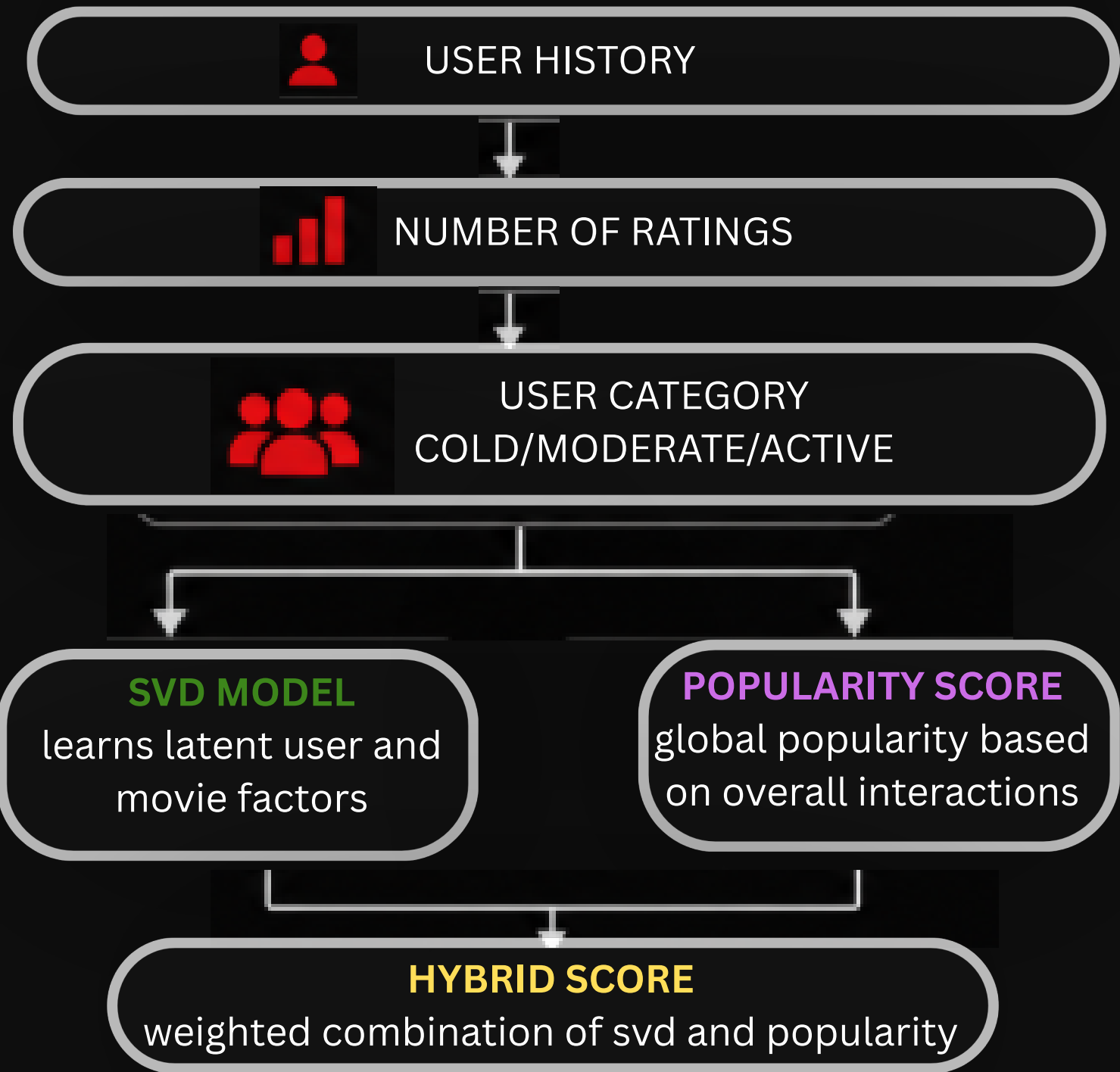
GOAL

Deliver accurate recommendations across all user types, including users with limited history.



We use SVD as the primary recommendation engine and dynamically incorporate popularity signals based on the user’s activity level

HOW IT WORKS



DYNAMIC WEIGHTING STRATEGY

USER TYPE	SVD WEIGHT	POPULARITY WEIGHT
 COLD USER (< 10 ratings)	75%	25%
 MODERATE USER (< 30 ratings)	90%	10%
 ACTIVE USER (≥ 30 ratings)	95%	5%

MATHEMATICAL FORMULA

$$\text{HybridScore} = w_{svd} \times \text{SVDScore} + w_{pop} \times \text{PopularityScore}$$

where, $w_{svd} + w_{pop} = 1$

WHY THIS APPROACH?

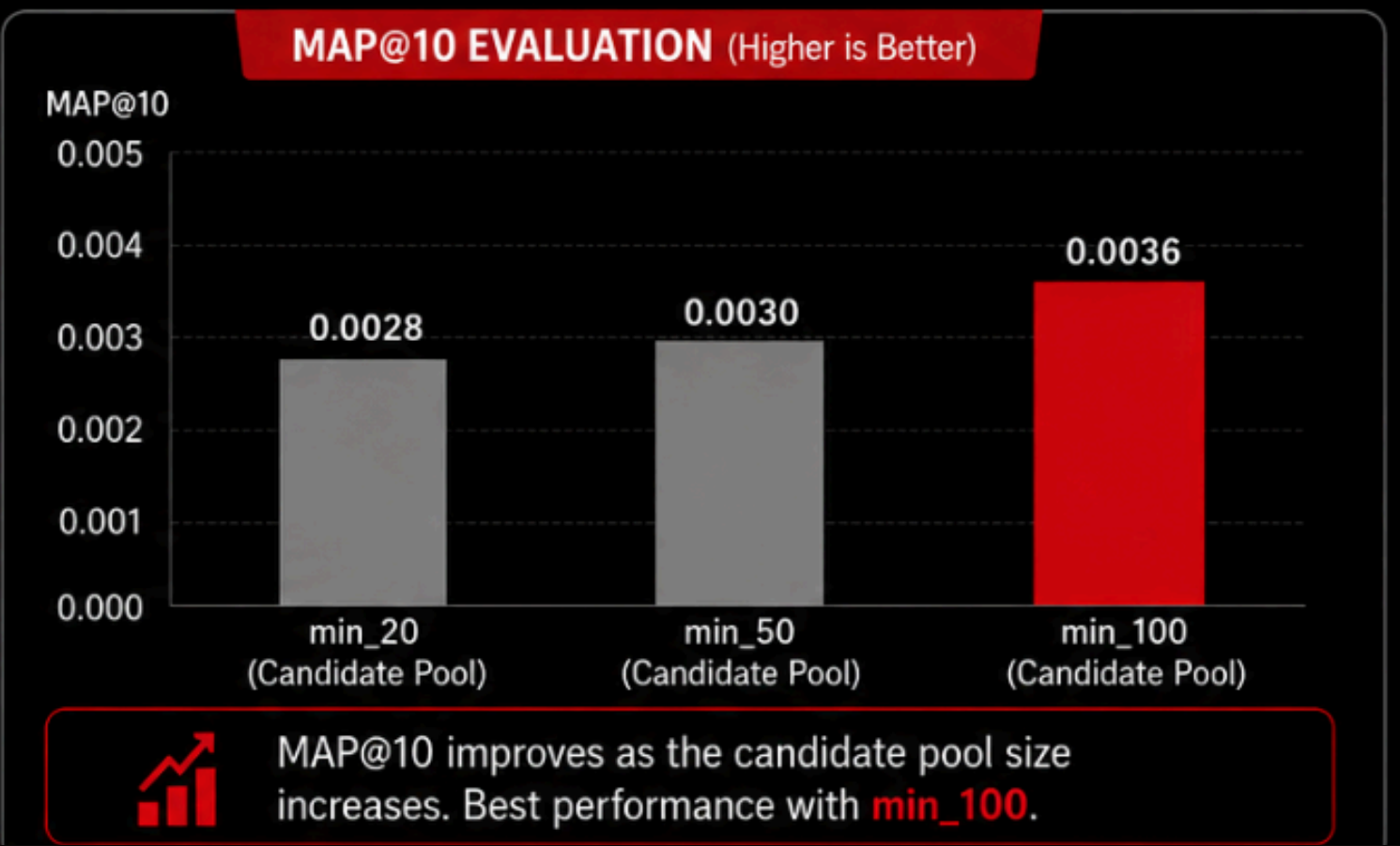
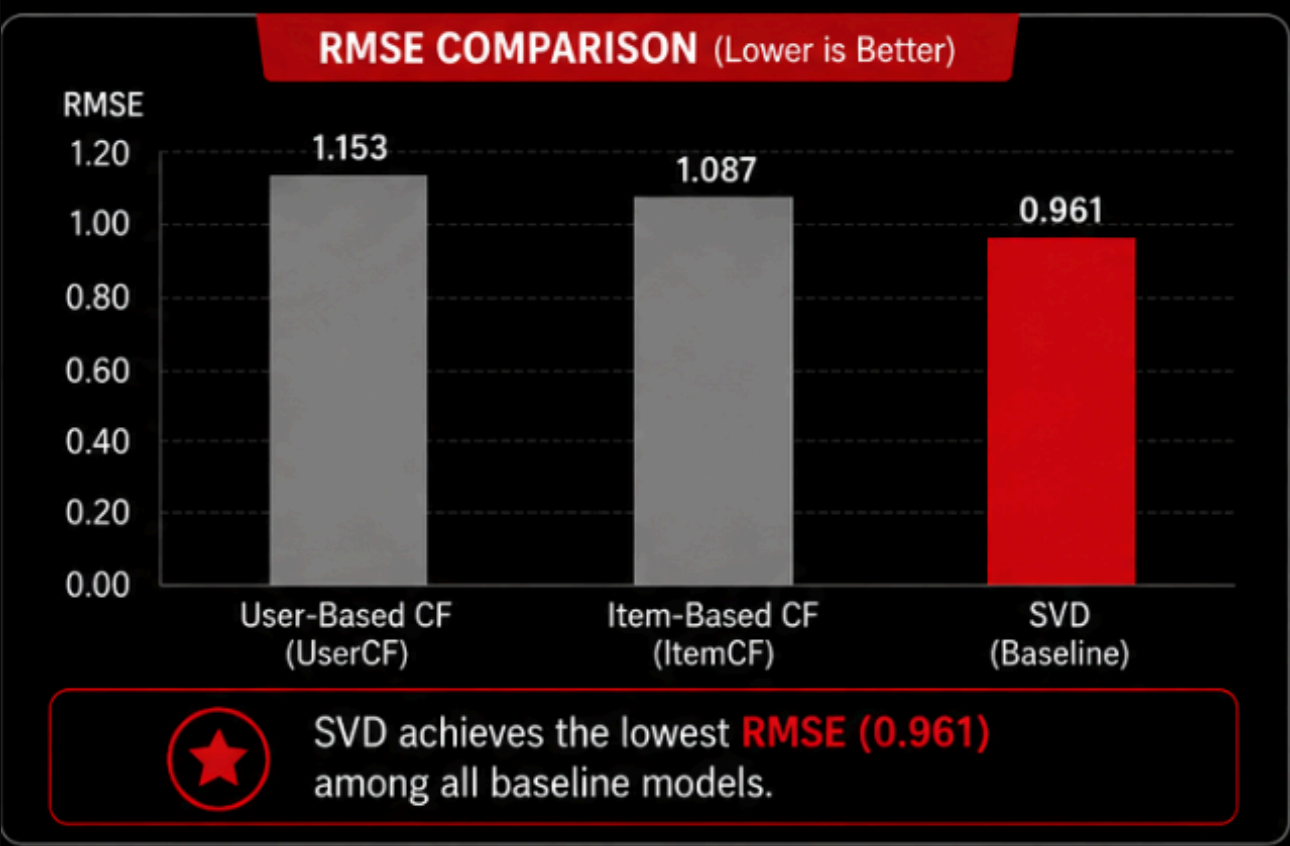


- ✓ Handles cold-start users effectively
- ✓ Preserves personalization through SVD
- ✓ Uses popularity when interaction history is limited
- ✓ Automatically adapts to user activity level

EXPECTED IMPACT



- ✓ More reliable recommendations for cold users
- ✓ Better personalization for active users
- ✓ Improved robustness across different user activity levels



MODEL COMPARISON		
Model	Strength	Weakness
 User-Based CF (UserCF)	Captures user preferences	Suffers in sparse data and cold-start cases
 Item-Based CF (ItemCF)	More scalable and stable	Less personalization for individual users
 SVD (Baseline)	Learns latent factors, best RMSE	Struggles with cold-start users

KEY FINDINGS



SVD achieved the lowest RMSE(0.961)



Matrix factorization outperformed neighbourhood-based methods



MAP@10 improved with larger candidate pools



Dynamic Hybrid SVD selected as the final recommendation model

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RECOMMENDATION EXAMPLES

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HEAVY USER (High Activity)



USER ID: 2439493



NUM RATINGS: 140

USER HISTORY (Highly Rated)	TOP RECOMMENDATIONS
The Jungle Book (1994) ★ 5	Finding Neverland (2004)
The Bells of St. Mary's (1945) ★ 5	The Shining (1980)
Bedknobs and Broomsticks (1971) ★ 5	Doctor Zhivago (2002)
Charlotte's Web (1973) ★ 4	Stargate SG-1: Season 7 (2003)
9/11 (2002) ★ 4	Stand By Me (1986)

COLD USER (Low Activity)

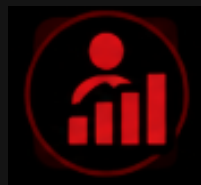


USER ID: 788



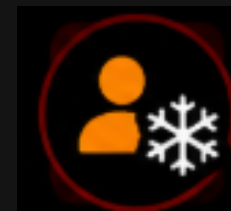
NUM RATINGS: 4

USER HISTORY (Previously Rated)	TOP RECOMMENDATIONS
Top Gun (1986) ★ 5	The Lord of the Rings: The Two Towers (2002)
Hitch (2005) ★ 4	The Lord of the Rings: The Fellowship of the Ring (2001)
Whale Rider (2003) ★ 4	The Princess Bride (1987)
Spider-Man 2 (2004) ★ 3	The Usual Suspects (1995)
	Raiders of the Lost Ark (1981)



HEAVY USER

Rich interaction history allows the model to learn stronger preference patterns, resulting in highly personalized recommendations



COLD USER

Limited interaction history reduces available preference signals. Popularity weighting helps generate reliable recommendations



FINAL TAKEAWAY

The Dynamic Hybrid SVD model adapts its recommendation strategy based on user activity, improving robustness across different user profiles



Netflix recommendation system

Netflix Prize Dataset · Dynamic Hybrid SVD Recommender

Total ratings

437,440

Users

68,716

Movies

13,984

Avg rating

3.55

Best RMSE (SVD)

0.961

MAP@10 (Hybrid)

0.0028

Controls

User ID

1

Recommendation model

Dynamic Hybrid SVD

Number of recommendations

10

How it works

1. Enter a User ID from the Netflix Prize dataset

2. Adjust Top-K recommendations

3. Adjust minimum popularity threshold

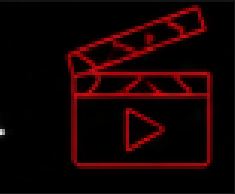
4. Click Get recommendations

Model Comparison

 **Dynamic Hybrid SVD**
Combines matrix factorization with popularity-based weighting to balance personalization and recommendation reliability across different users.

 **Cold-Start Handling**
Adapts recommendation weights based on user activity, improving results for users with limited rating history while maintaining personalization.

 **Interactive Dashboard**
Provides a user-friendly interface for generating recommendations, exploring model outputs, and analyzing system performance in real time.



CONCLUSION

The proposed Dynamic Hybrid SVD model successfully combines personalization and popularity-based ranking to generate robust recommendations across different user activity levels.